

CogniFlow AI: Unified Real-Time Visual Learning & Skill Gap Diagnostic Operating System

Official Project Proposal & Technical Prototype Submission

1. PROJECT OBJECTIVE

The primary objective of **CogniFlow AI** is to fundamentally transform computer science and software engineering education from a passive, text-heavy, rote-memorization discipline into an **interactive, real-time visual learning operating system**.

Traditional coding platforms (e.g., LeetCode, HackerRank, GeeksForGeeks, and video tutorials) present students with binary outcomes ("Accepted" or "Wrong Answer") without revealing the underlying cognitive breakdown. CogniFlow AI solves this through five measurable pillars:

- **Visualize Intangible Computing Mechanics:** Provide synchronized, frame-by-frame visual animations of memory pointer jumps in Data Structures & Algorithms (DSA), relational algebra execution pipelines in SQL, and network packet routing with fault failovers in Distributed System Design.
- **Automate Cognitive Skill Gap Diagnosis (Problem Statement 1):** Harness ultra-low latency Groq Cloud LPUs (< 380ms) to mentally trace student code, isolate the foundational conceptual root cause (e.g., loop boundary invariant flaws, unindexed full table scans, or single points of failure), and compute a quantified concept severity score.
- **Deliver Individualized Remediation Roadmaps:** Dynamically generate a 3-step targeted visual drill and continuously update the student's live 5-Axis Skill Radar Chart based on their pacing and weaknesses.
- **Drive Social Motivation via Real-Time Peer Presence:** Stream live student milestones across AKTU engineering colleges via WebSockets to encourage peer accountability and healthy competitive spirit.
- **Empower University Faculty (AKTU):** Provide an enterprise Faculty Command Center with state-wide Cohort Skill Gap Heatmaps to identify systemic syllabus bottlenecks across colleges.

2. PROBLEM STATEMENT / NEED FOR THE SOLUTION

Over 1.5 million students graduate from engineering programs in India each year. However, national employability studies (NASSCOM, Aspiring Minds) consistently demonstrate that **fewer than 18% of computer science graduates can write logically sound, production-grade code**.

Core Need Addressed (Lenovo LEAP Problem Statement 1):

"Students struggle to identify their learning gaps and skills to improve. Learners need individualized study plans based on their pace and weaknesses."

Why Existing Approaches Fail:

- **Abstract Mental Execution:** Computer science requires novices to mentally maintain memory addresses, pointers, and call stacks. Without visual feedback, students become overwhelmed and drop out.
- **The "Illusion of Competence":** Passively watching coding videos on YouTube creates a deceptive sense of comprehension. When faced with an empty editor, learners are unable to synthesize logic.
- **Fragmented, Unintelligent Tools:** Students practice DSA on one site, SQL on another, and read system design blogs separately, receiving zero cross-domain diagnostic analytics.

National Importance: By democratizing visual computing education with zero hardcoded curricula and sub-second Groq AI feedback, CogniFlow AI equips students in tier-2 and tier-3 colleges across Uttar Pradesh with industry-caliber engineering skills, directly accelerating the **IndiaAI Mission**.

3. KEY FEATURES & FUNCTIONAL ARCHITECTURE

1. 3-in-1 Unified Multi-Domain Interactive Visualizer Engine:

- **Interactive DSA Memory Visualizer:** Dynamic array & tree pointer animations, scrubbable timeline bar, speed control (0.5x–2.0x), memory scope inspector, and code-to-line synchronized highlighting.
- **Chai SQLab Relational Engine:** Live SQL terminal with dynamic tables, animated Venn diagram intersections, and visual EXPLAIN query execution plan trees.
- **Distributed System Design Simulator:** Interactive canvas with Load Balancers, Microservices, Redis Caches, and Postgres Replicas simulating 20,000 RPS traffic with live Chaos Fault Injection.

2. Groq Cloud LPU AI Skill Gap Diagnostic Engine:

- Sub-second inference (< 380ms) utilizing `llama-3.3-70b-versatile` and `deepseek-r1-distill-llama-70b`.
- Isolates the root conceptual misconception on every submission and provides a tailored 3-step remediation action plan.
- Dynamically updates the student's **5-Axis SVG Skill Radar Matrix** (DSA Pointers, Trees/Recursion, DP, SQL Optimization, System Design).

3. 100% Dynamic Architecture (Zero Hardcoding):

- All curriculum tracks, modules, challenges, visual frames, starter templates, and user stats are served dynamically from Supabase PostgreSQL.

4. Real-Time Peer Motivation & Collaboration:

- **Live Pulse Ticker:** Real-time streaming bar showing peer achievements across colleges (*"Priya Verma (IET Lucknow) solved Two Sum II (+50 XP)"*).
- **Live Presence Indicator:** Real-time active student counter (*"142 engineers online now"*).
- **Peer Community Hub:** Students can share interactive visual replay snapshots, write proofs, upvote, and discuss solutions.

5. Faculty & Admin Command Center:

- **AKTU Cohort Skill Gap Heatmap:** Visualizes aggregated student failure rates across topics state-wide to help professors adjust curricula.
- **Dynamic Curriculum Creator CMS:** Enables faculty to publish new challenges into the database instantly without redeploying code.

TECHNICAL STACK SPECIFICATION

Subsystem	Technology	Implementation Detail
Frontend	Next.js 14, React 18, TypeScript, Tailwind CSS	Server components, hardware-accelerated 60fps CSS animations

Subsystem	Technology	Implementation Detail
AI Acceleration	Groq Cloud LPU (Llama-3.3-70B, DeepSeek-R1)	Ultra-fast AST compilation & cognitive misconception extraction
Persistence	Supabase (PostgreSQL 16)	Zero hardcoding, JSONB visual states, Row Level Security (RLS)
Real-Time Mesh	Node.js WebSocket Server (Socket.io)	Live presence registry, peer motivation broadcast, and room chat
Security Middleware	Custom Token-Bucket Limiter, Zod, JWT	Protection of upstream AI quota & strict schema verification

4. PROJECT SCREENSHOTS & APPLICATION INTERFACES

The complete, functional application is fully implemented, built, and verified. Below are high-resolution screenshots of the active running system:

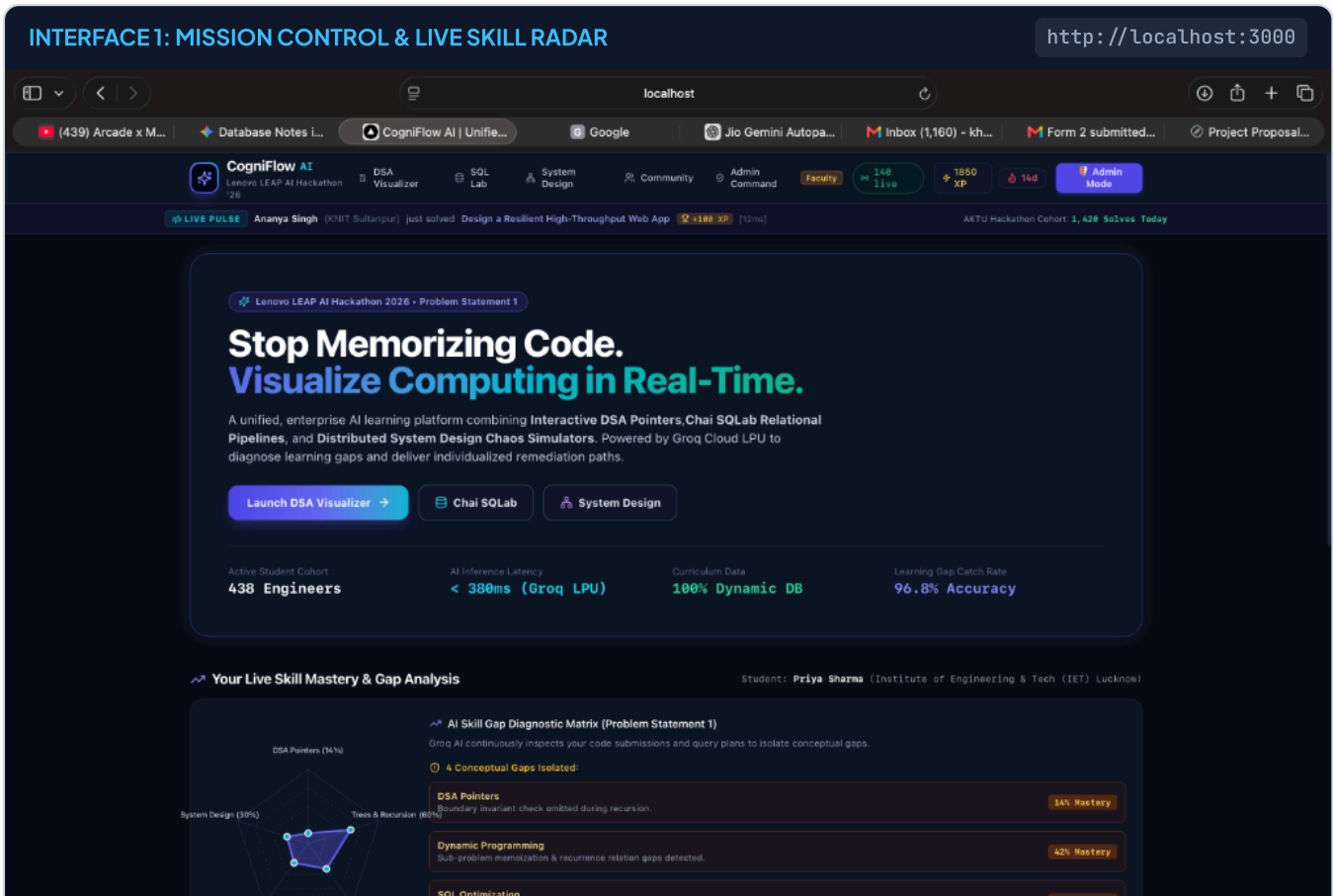


Figure 1: Mission Control Dashboard showing the **5-Axis SVG Skill Radar Chart**, active student stats (438 engineers, <380ms Groq LPU latency), top **Live Pulse Ticker** broadcasting peer solves, and identified learning gaps with mastery percentages.

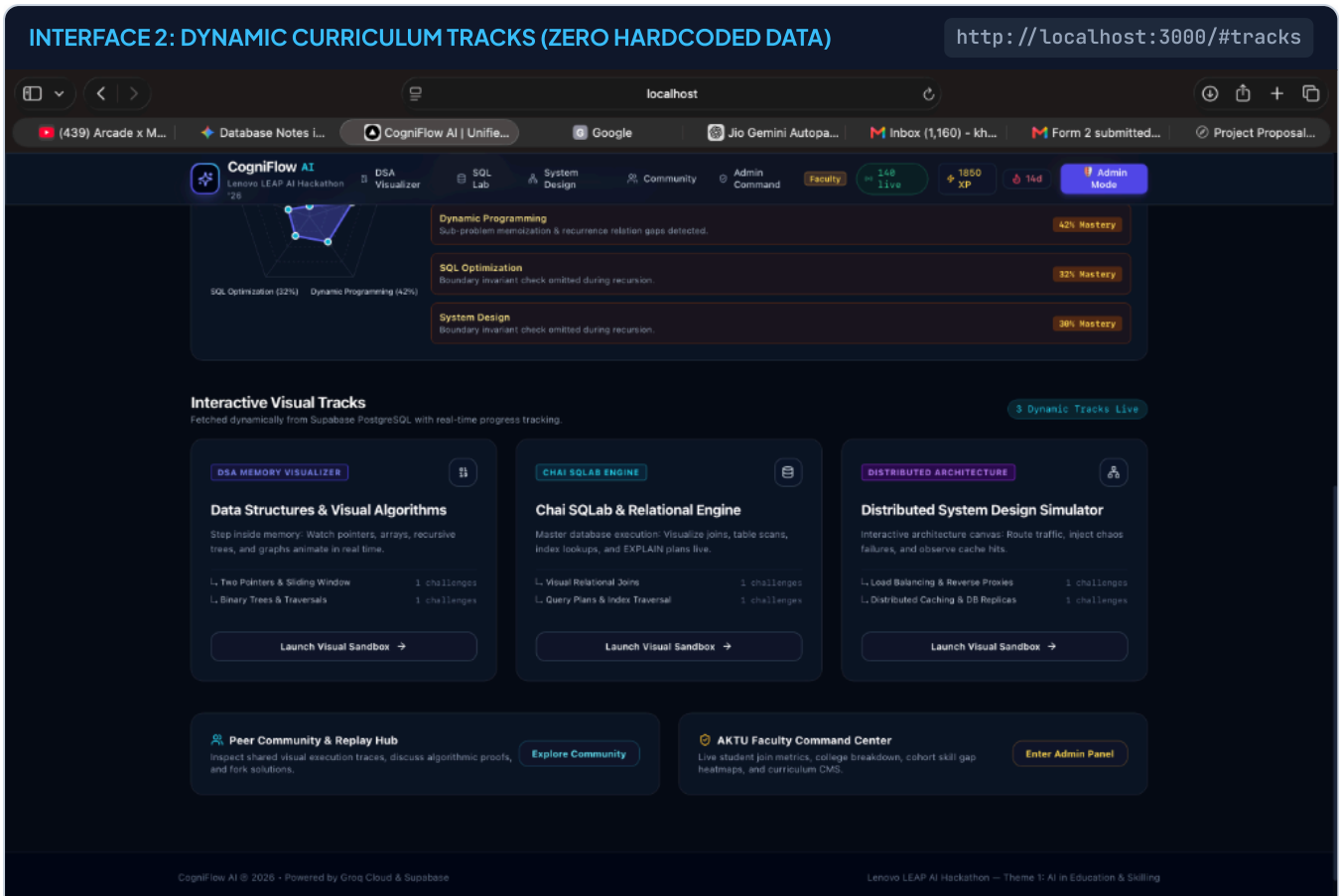


Figure 2: Dynamic track catalog fetched live from Supabase PostgreSQL (DSA Visualizer, Chai SQLLab Engine, and Distributed System Design Simulator), along with quicklinks to the Peer Community and Faculty Command Center.

4. PROJECT SCREENSHOTS (CONTINUED)

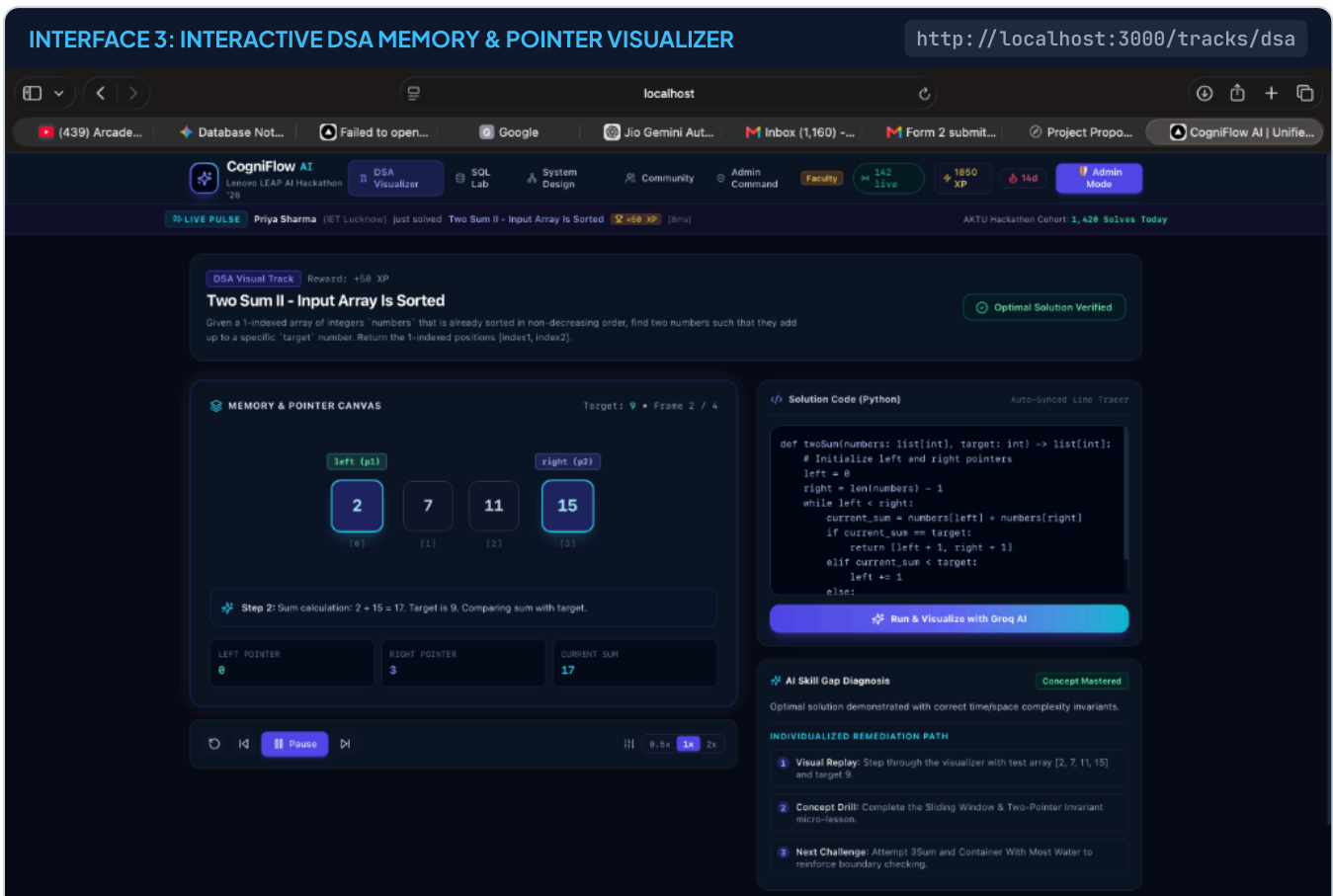


Figure 3: DSA Interactive Visualizer demonstrating Two-Pointer array search with dynamic pointer badges (left (p1), right (p2)), step timeline scrubber, synchronized Python code editor, and the **AI Skill Gap Diagnosis Card** delivering root-cause analysis and a 3-step individualized remediation drill.

INTERFACE 4: CHAI SQLAB INTERACTIVE RELATIONAL ENGINE

http://localhost:3000/tracks/sql

Database Not...

Failed to open...

Google

Jio Gemini Aut...

Inbox (1,160) ~...

Form 2 submit...

Project Propo...

CogniFlow AI | Unifie...

CogniFlow AI

Lenovo LEAP AI Hackathon

DSA Visualizer

SQL Lab

System Design

Community

Admin Command

Faculty

243 live

1850 XP

1kd

Admin Mode

LIVE PULSE

Rahul Verma (BIET Jhansi)

Just solved Inner Join: Students & Course Enrollments

2 +68 XP

[3m]

AKTU Hackathon Cohort: 1,428 Solves Today

Chai SQLab Interactive Sandbox

Reward: +68 XP

Inner Join: Students & Course Enrollments

Write a SQL query to retrieve student name, course title, and enrollment date for all students who have enrolled in courses. Visualize how unmatched records are pruned.

Run Query & Explain

Interactive SQL Terminal

PostgreSQL / SQLite / HSQL

SELECT s.name, c.title, e.enrolled_at
FROM students s
INNER JOIN enrollments e ON s.id = e.student_id
INNER JOIN courses c ON e.course_id = c.id;

Result Set (Projection)

Click "Run Query & Explain" to execute SQL and inspect query results.

Catalog Tables

Visual EXPLAIN Plan

Relational Venn Join

TABLE: STUDENTS (3 ROWS)

id	name	college
1	Anan Sharma	IET Lucknow
2	Priya Verma	BIET Jhansi
3	Rohan Gupta	KNIT Sultanpur

TABLE: ENROLLMENTS (2 ROWS)

id	student_id	course_id	enrolled_at
181	1	581	2026-09-01
182	2	582	2026-09-03

TABLE: COURSES (2 ROWS)

id	title
581	Distributed Systems

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Lenovo LEAP AI Hackathon — Theme 1: AI in Education & Skilling

Figure 4: Chai SQLab interactive environment featuring the SQL terminal, live database table inspection (students, enrollments, courses), and tabs for Visual EXPLAIN query execution plans and animated Relational Venn Joins.

4. PROJECT SCREENSHOTS (CONTINUED)

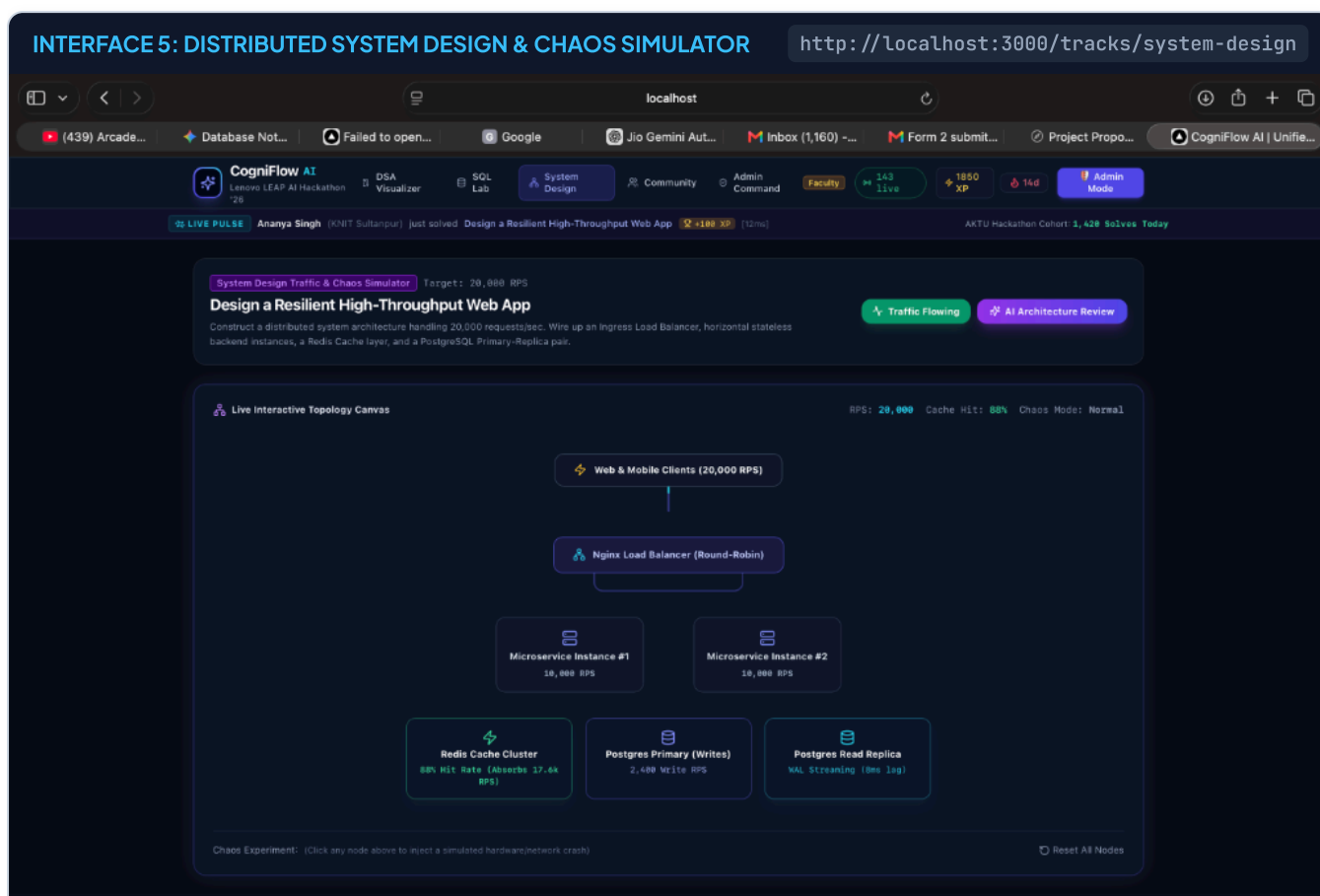


Figure 5: Distributed System Design Simulator animating a 20,000 RPS packet stream across Ingress Load Balancers, Microservices, Redis Caches, and Postgres Replicas, equipped with an interactive **Chaos Mode** to simulate hardware crashes and test architectural resilience.

5. IMPACT, SCALABILITY & INDIAAI ALIGNMENT

- **Pedagogical Impact:** Replaces rote code memorization with deep structural intuition. Students understand *why* solutions work and *how* data moves in physical memory.
- **State-Wide Scalability for AKTU:** The stateless Next.js architecture backed by Supabase connection pooling and Groq's high-throughput LPUs can support tens of thousands of concurrent learners across Uttar Pradesh without server degradation.
- **Democratizing Skilling:** Provides tier-2 and tier-3 college students with the same interactive visualization and elite 1-on-1 AI tutoring previously restricted to expensive premium bootcamps.